AI SKILL TREE 2026

www.arianprabowo.com

Want to get better at AI, but don't know what you don't know? From daily chat (L1) to autonomous coding agents (L3–L4) and IMO-math / cyber-RL (L6), here is the full progression map.

① DIAGNOSE: Scan down each column (L1 → L6) to find where you stop
② LEVEL UP: Target the Top 3 pills in the row below
③ UNLEARN: Avoid the OBSOLETE traps on the right

*Scope: Modern LLMs, Multimodal (mmLLM) & Agentic AI across end-users, MLE & model research (not GOFAI or classic ML/DL). Changes fast — see the AI Ladder Evolution mega-chart by Arian Prabowo · Updated: 2026-11-07 17:58:53 AEDT (UTC+11:00)

AI LEVEL	THINK, SEARCH & ANALYZE	CREATE & DESIGN (MEDIA)	AUTOMATE & BUILD (CODE)	OBSOLETE
Overview · Persona · Surface · Use Case	Research · Data & Docs · RAG · Reasoning & Alignment	Image · Video · Audio · Voice · 3D · Provenance & DiT	Workflows · AI IDEs · Agents · Evals, Safety & Training	Still Doing This? You're Behind
LEVEL 1 **EVERYDAY CONSUMER** Uses default consumer AI apps for conversational search, writing, media, and zero-install web mini-apps. • Persona: Casual daily user & general public. • Surface: Phone app, web browser & built-in OS assistant on any phone or laptop. • Use Case: Search replacement, drafting emails/texts, summarizing PDFs, live voice & browser mini-apps.	**ChatGPT / Claude Daily Q&A** **Perplexity Cited Web Search** **PDF & Article Upload Summary** Google Search AI Overviews · Gemini Mobile Assistant Web Browsing in Chat · Android Circle to Search Apple AI Writing Tools · Meta AI in WhatsApp / IG Hallucination Fact-Check · Draft & Rewrite Emails DeepSeek Free Web / App · TLDR Thread Summarize · Grammar & Tone Polish YouTube Video Summary · Gmail / Outlook Auto-Draft · Foreign Language Chat Travel Itinerary Planner · Excel / Sheets Formulas · Click Cited Source Links Memory On / Off Toggle · Data Training Opt-Out · Temporary Incognito Chat	**ChatGPT Live Voice Mode** **Conversational Image Gen (4o)** **Phone Camera Live Vision Q&A** Gemini Live Voice Chat · Google Lens Visual Q&A Mobile OS Voice Dictation · Magic Eraser Photo Unblur Canva Magic Media Studio · iOS Genmoji / Image Wand Screenshot OCR Translate · Zoom / Meet AI Summaries Google Imagen / ImageFX · Bing Image Creator · Auto-Captions on Video Spotify AI Playlist DJ · 1-Click BG Removal · Receipt Photo Expense Voice Memo Transcript · Google Photos Ask-Photos · Whiteboard Photo Read Menu Photo Translation · Spot AI / Deepfake Tells · Short-Video Auto-Clip	**Claude Artifacts (Web App / SVG)** **ChatGPT Canvas Live Preview** **Zero-Install Browser Vibe-Code** Single-File HTML/JS Mini-App · Interactive React Calculator Gemini Canvas Web Preview · Shareable Artifact App Link Browser Quiz & Flashcard App · Interactive Chart & Dashboard 1-Click Copy & Run Code · Fix Error by Pasting Message Mermaid Flowchart in Chat · SVG Diagram & Logo Code · Browser Mini-Game (Snake/2048) Split-Screen Code Preview · Ask AI to Explain Code · Convert Table to Web App Interactive Budget Tracker · Landing Page Mockup in Chat · Client-Side Only (No Keys) Download Single .html File · Iterative 'Make Button Blue' · Sandbox Preview Reset	**"Act as a..." Jailbreaks** **Buying 10k Prompt PDFs** **Blindly Trusting AI Claims** ChatGPT w/o Web Search · Copy-Pasting 1 Para Only ChatGPT Plugins Store · Google Bard / Bing Sydney DALL-E mini / Craiyon · Short 3-Word Prompts Passwords in Public Chat · AI Voice-Clone Scams AI as Medical/Legal Fact · Ignoring Source Links
LEVEL 2 **WORKSPACE POWER USER** Orchestrates paid reasoning models, deep-research agents, and interactive canvases across multi-file workspaces. • Persona: Knowledge worker, analyst, designer & solo operator. • Surface: Paid web workspaces, desktop apps, multi-source notebooks & creative studios. • Use Case: Deep research reports, multi-doc synthesis, interactive canvases & consistent media.	**Multi-Step Deep Research** **Extended Thinking (o3 / Claude)** **Data Analysis & Chart Canvas** Projects & Custom GPTs · NotebookLM 50-Source Pod DeepSeek-R1 DeepThink · Gemini 1M–2M Token Upload Perplexity Pro Spaces · Multi-Doc Cross-Audit Zero-Data-Retention (ZDR) · Critique-Then-Revise Loop Grok-3 DeepSearch Mode · LMSYS Chatbot Arena · Side-by-Side Model Check Context Token Trimming · CSV / Excel Code Sandbox · PII Redaction Before Upload Full-Book PDF Q&A · Earnings Call Synthesis · Literature Review Matrix Rubric Self-Grading · Role-Separated Debate · Citation Source Audit	**Midjourney --sref / --cref Refs** **Sora 2 / Veo 2 / Runway Video** **ElevenLabs Instant Voice Clone** NotebookLM Audio Host · Kling / Luma Keyframes Ideogram / 4o Text-Image · Suno v4 / Udio Song Gen HeyGen Video Translate · Superwhisper / Wispr Flow Adobe Firefly Gen-Fill · Live Screen-Share Vision Pika Scene Ingredients · Descript Overdub & Eye · Topaz AI Video Upscale Captions.ai Avatar · Hedra Character-2 Sync · Leonardo Phoenix Canvas Recraft Brand Vectors · Spline AI 3D Prompts · Stem Splitter Isolation Photoshop Gen-Expand · Commercial IP-Safe Models · Voice Consent Verification	**Lovable / Bolt / v0 Web Apps** **ChatGPT Agent / Operator Mode** **Zapier / Make No-Code AI Flows** Replit Agent / v0 Templates · Scheduled AI Tasks & Alerts Notion AI & M365 Copilot · Raycast AI Quick Commands Granola / Fireflies Notes · Airtable / Sheets AI Agents Custom GPT Action Webhooks · Workspace SSO & Audit Logs Apple Shortcuts AI Actions · Browser AI Sidebar Actions · Linear / Jira AI Triage Superhuman / Gmail AI Rules · Slack AI Channel Digests · Loom AI Auto-Workflows Tally / Typeform AI Routing · Coda AI Doc Automations · TextExpander / Alfred Snips Readwise Reader AI Ghost · Mermaid / Whimsical AI Flow · Role-Based Access (RBAC)	**"Let's think step by step"** **"Prompt Engineer" ($300k)** **Uploading Client PII to Chat** Tipping $200 / Threats · "6 fingers" Neg Prompts Midjourney Discord Bot · Custom GPT Store Hype Manual 4k Text Splits · 1-Page Prompt Walls Copy-Pasting CSV Tables · Manual Zoom Note-Taking Fake Legal Citations · Shadow AI w/o IT SSO
LEVEL 3 **VIBE CODER & BUILDER** Ships full-stack apps and multi-step automations by directing AI IDEs, CLI coding agents, and cloud APIs. • Persona: Indie builder, automation maker & AI-native developer. • Surface: AI IDEs (Cursor/Windsurf), CLI agents (Claude Code), app builders & APIs. • Use Case: Natural-language app shipping, multi-file repo refactoring, MCP/webhook flows & API media pipes.	**Spec-Driven Dev (Spec Kit / PRD)** **Tavily / Exa / Firecrawl API** **Repomix / Gitingest Packing** System vs User Role Split · llms.txt Docs Standard Jina Reader URL-Markdown · Context Window Budgeting Markdown Memory Banks · Multi-Step Prompt Chains Few-Shot JSON Examples · OpenRouter Benchmarking Perplexity Sonar API · Brave Search Grounding · Unstructured PDF API Prompt Git Versioning · Token Cost & Spend Caps · Fallback Model Retries Streaming SSE Responses · Temp / Top-P API Tuning · Stop-Sequence Control Needle-in-Haystack Check · Untrusted Input Delimiters · SQL / Pandas Query via AI	**ComfyUI Ready Workflows** **Fal.ai / Replicate Webhooks** **ElevenLabs / Vapi Voice API** Black Forest Flux API · Remotion Code-to-Video Whisper + FFmpeg Subs · Sync.so Lip-Sync API Recraft v3 Vector API · Figma-to-Code Plugins Magnific / Krea Upscale · Luma / Meshy 3D Assets Runway Act-One API · Hume Empathic Voice API · Retell Phone Webhooks AssemblyAI Diarization · Creatomate JSON Video · Photoroom Batch API LivePortrait Face API · Mux Video Auto-Chapters · PlayHT Low-Latency TTS Gladia Multilingual ASR · C2PA / SynthID Metadata · NSFW Moderation API Gate	**Cursor / Windsurf Agent Mode** **Claude Code CLI & CLAUDE.md** **SKILL.md & Subagent Workflows** .cursor/rules & AGENTS.md · Claude Desktop MCP & n8n Context7 Live Docs MCP · Plan-Before-Act & Diffs Human-in-Loop Tool Approve · Codex CLI / OpenCode / Aider Browser-Use / Stagehand · Generative UI (Vercel SDK) Ollama / LM Studio Local · Groq / Cerebras LPU API · LiteLLM / CF AI Gateway Together / Fireworks API · Helicone / LangSmith Logs · OWASP AI Code & Rules Check Red-Green TDD & E2E Tests · Secret & API-Key Scrubbing · Regex / Bash / Git via AI GitHub Actions PR Bot · Slopsquat Package Check · Stripe + Auth0 Vibe Ship	**AutoGPT / BabyAGI Loops** **Agent Deleting Failing Tests** **Blindly Running Agent rm -rf** API Keys in Public Git · Poisoned Rules / Issue Text Legacy .cursorrules Dump · 50-Node Flowise Chains Manual Regex Scraping · Pasting Files to Web UI No Git Commit Pre-Agent · Shipping AI SQL w/o Auth No API Spend / Rate Cap · Fake Slopsquat Packages
LEVEL 4 **AI & AGENT ENGINEER** Architects production multi-agent backends with schema-enforced tool use, hybrid RAG, guardrails, audit traces, and CI evals. • Persona: Production AI software engineer & systems architect. • Surface: Python/TS SDKs, custom MCP servers, state graphs, microVM sandboxes & observability stacks. • Use Case: Stateful multi-agent backends, hybrid RAG, guardrails, audit trails, compliance & CI eval gates.	**Pydantic / Zod Schema Outputs** **Hybrid RAG + Cross-Rerankers** **ColPali & Docling PDF RAG** Prompt Caching (90% Saved) · Hallucination & Faithfulness Anthropic Thinking Tokens · Contextual & GraphRAG DSPy Prompt Optimization · BAML Typed Prompt Calls Cohere Rerank-v3.5 Pipe · Citation Span Grounding Hypothetical Embed (HyDE) · Parent-Doc Late Chunking · Reciprocal Rank Fusion Agentic RAG & Search-R1 · Semantic Intent Router · Multi-Query RAG Expand Surya / LayoutLM Tables · Demographic Fairness Eval · PII & Bias Leakage Audits Faithfulness Gate Check · Sleep-Time / Dream Memory · Golden Retrieval Recall	**OpenAI / Gemini WebRTC API** **LiveKit / Pipecat Voice AI** **Programmatic ComfyUI API** Deepgram Nova-3 Stream · Cartesia / Kokoro TTS Vision-LLM UI Grounding · Barge-In VAD Interruption Twilio SIP Call Bridging · TwelveLabs Video Search CLIP / SigLIP Visual RAG · WebSocket PCM16 Streams Frame-Accurate Video Cut · Silero VAD Turn-Taking · Daily.co WebRTC Mesh HeyGen Streaming Avatar · WhisperX Word Timestamps · pyannote Speaker Diarize Bounding-Box Detection · Deepfake / Spoof Detection · C2PA Cryptographic Sign Sub-500ms Voice Loop · Audio Watermark Check · 3D Gaussian Asset API	**Custom MCP Servers & OAuth** **LangGraph Durable State Graphs** **Promptfoo / Braintrust CI Evals** E2B / Bubblewrap Sandboxes · Audit Trails & OTel Traces Llama-Guard / NeMo Rails · LLM-as-a-Judge Rubrics Recursive Harness (Harness²) · OWC Triad & Loop/Graph Eng Qdrant / Milvus / pgvector · SWE-bench & Terminal-Bench PyRIT / Garak Red-Teaming · EU AI Act / SOC2 Audit Logs · Harness Behavior-Code Audit Human-in-Loop Interrupts · Agent Memory (Mem0/Letta) · Temporal Durable Exec Redis Semantic Caching · Langfuse / Phoenix Spans · Cascade Model Cost Router Canary / Shadow Deploys · Confused-Deputy & RBAC Gate · A2A / ACP & Agent Gateways	**Shipping w/o Evals or Traces** **Naive 500-Tok Vector RAG** **Regex-Parsing LLM JSON** Un-Sandboxed Code Exec · Fine-Tuning to Add Facts LangChain LLMChain Class · Assistants API v1 Polls text-embedding-ada-002 · Pinecone Pod-Index Lock Unicode Tag / ANSI Smuggling · No Tool Audit Logging God-Node / Unbounded Loops · Vibe-Check Only Evals
LEVEL 5 **LOCAL & POST-TRAINING** Serves quantized open-weight models on dedicated GPUs/Macs and aligns weights via QLoRA, DPO, GRPO, and steering. • Persona: ML systems engineer, local-first builder & model post-trainer. • Surface: Multi-GPU rig (RTX 3090/4090/5090), unified-memory Mac, vLLM, SGLang & Unsloth. • Use Case: High-throughput local serving, QLoRA/DPO/GRPO alignment, weight merging & air-gapped AI.	**Synthetic CoT Dataset Curation** **Rejection-Sampled SFT / DPO** **Grammar-Constrained GBNF / JSON** Local DeepSeek-R1 & Qwen-3 · Constitutional / RLAIF Data RoPE Scaling (YaRN) · Min-P / DRY / XTC Sample KTO / IPO / SimPO Alignment · Custom ChatML Templates Air-Gapped Hybrid RAG · Domain Eval Benchmarks Refusal Boundary Tuning · Bias & Toxicity Filtering · Perplexity Quant Audit KL-Divergence Drift Check · Continued Pretraining (CPT) · ShareGPT Multi-Turn Data System Prompt Distill · Local BGE-Reranker · Air-Gapped Citations CoT Trace Length Tuning · RAGAS Local Faithfulness · Data Contamination Check	**ComfyUI Custom Node Pipelines** **Flux.1 / Wan-2.1 Local Weights** **Kohya / AI-Toolkit LoRA Train** ControlNet & IP-Adapter · Whisper-v3-turbo Local Kokoro-82M / F5-TTS Local · RVC / So-VITS Voice Clone Florence-2 & SAM-2 Vision · GGUF FP8 Flux Offload TensorRT Diffusion Accel · HunyuanVideo / Mochi Run TeaCache / WaveSpeed · Chatterbox / Orpheus TTS · faster-whisper CTranslate AnimateDiff Motion Nodes · Depth-Anything ControlNet · InstantID / FaceID Nodes CosyVoice2 / Fish-Speech · Ultimate-SD Tiled VAE · SageAttention Speedup JoyCaption VLM Tagging · Concept Erasure LoRA · Watermark-Preserving Gen	**vLLM / SGLang GPU Serving** **Unsloth / Axolotl / TRL QLoRA** **DPO / ORPO / Local GRPO Tune** EXL2 / AWQ / GGUF / FP8 · SGLang RadixAttention Outlines / XGrammar FSM · Speculative Decode (EAGLE) DoRA / rsLoRA / PiSSA / IA3 · Activation Steering Vectors KV-Cache FP8 / INT4 Quant · Apple MLX / Dual-4090 Rig llama.cpp / KoboldCpp · MergeKit SLERP/TIES/DARE · vLLM Multi-LoRA Serving Completion-Mask Packed SFT · Machine Unlearning Runs · Local Stdio MCP & SQLite Hermes-3 Local Tool Call · OmniParser GUI Grounding · GCG / AutoDAN / BoN Red-Team Prefill-Decode Disagg · Zeroth-Order (MeZO) Opt · imatrix Importance Quant	**Full-Weight 7B Style FT** **Automatic1111 SD 1.5 UI** **SFT Without KL / Eval Checks** 8-bit bitsandbytes Slow · Oobabooga Text-Gen UI Manual GPTQ Calibration · DreamBooth 40m Colab LoRA Rank-128 Overfit · Raw Alpaca 52k SFT Set Static RoPE Alpha Hacks · Flask Single-Batch API Blind Frankenmerges · FT Without Safety Evals
LEVEL 6 **FRONTIER RESEARCHER** Pre-trains foundation models, designs verifiable RL & scalable-oversight alignment, and authors custom GPU kernels. • Persona: Foundation model scientist, alignment researcher & kernel author. • Surface: Multi-node H100/B200/TPU clusters, InfiniBand fabrics, FSDP2/Megatron & CUDA/Triton. • Use Case: Base/MoE pre-training, RLVR/GRPO, mechanistic interpretability, scalable oversight & GPU kernels.	**RLVR (Verifiable Reward RL)** **GRPO / DAPO / Dr. GRPO RL** **Test-Time Compute Scaling** Scalable Oversight & Debate · Reward Hacking Defense Process Reward Model (PRM) · MCTS Tree-Search Reason AlphaProof & OpenMath Lean · Deliberative CoT Alignment Pass@k → Pass@1 Distill · Benchmark Decontam LSH Scheming & Sandbag Evals · Weak-to-Strong Generalize · Instruction Hierarchy RL Aha-Moment Backtracking · VinePPO / ReST-MCTS* Tree · KL-Divergence Ref Clip Curriculum Difficulty RL · Coconut Latent Space CoT · Math-Shepherd Verifiers CoT Monitorability Checks · RLCR Calibration & Collapse · CMA-ES & AlphaEvolve RL	**Rectified Flow Matching (DiT)** **Native Early-Fusion Omni-Modal** **3D Spatio-Temporal Video VAE** Discrete Audio Tok (Mimi) · Consistency Flow Distill Autoregressive Visual Tok · Interleaved Omni Attend Chameleon / Transfusion · 3D Gaussian Splatting Full-Duplex Speech Model · JEPA / Genie-2 World Sim Cross-Modal Contrastive · Rotary 3D-RoPE Video Pos · CFG Guidance Distillation MMDiT Dual-Stream Blocks · VQ-GAN / FSQ Scalar Quant · MAR Masked Autoregressive SoundStorm Parallel Audio · Latent Concept Erasure · Tree-Ring Latent Watermark Spatio-Temporal Patchify · Distribution Match Distill · Physics Video Priors	**FSDP2 / Megatron 3D Parallel** **Custom Triton & FlashAttention-3** **DeepSeek MLA & Aux-Free MoE** Mech Interp & SAE Circuits · FP8 / MXFP8 Mixed Train Exec-Guided Code RLVR · XBOW / Cybench Zero-Day RL ThunderKittens / CUTLASS · torch.compile & Inductor InfiniBand & NCCL Tuning · OpenRLHF / VeRL Clusters ASL / CBRN Frontier Evals · Constitutional Classifiers · Chinchilla / MuP Scaling DualPipe Compute Overlap · Multi-Token Predict (MTP) · WSD Warmup-Stable-Decay Ring-Attention 1M+ Context · FlexAttention & Pallas · NVSHMEM All-to-All MoE Micro-Scaling NVFP4 · Zero-Bubble Pipe Parallel · Harness Distillation (RL)	**4-Model PPO RLHF Pipeline** **Dense-Only 500B+ Pretrain** **Unverifiable Code/Math RMs** Scaling w/o Safety Evals · nn.DataParallel / DDP Static Sinusoidal Pos · FP32 Unfused Attention GANs for Image / Video · Dropless MoE Aux-Loss Human-Only RL Oversight · Contaminated Benchmarks Unmonitored Latent CoT · Length-Biased RMs